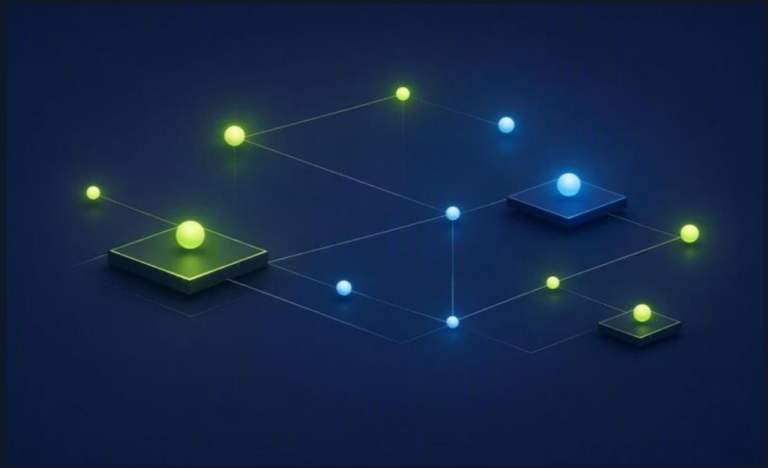


Next-Gen Enterprise AI Architecture

Moving Beyond Vendor Lock-In: The “Buy vs. Assemble” Debate

The Reality of the Enterprise AI Landscape

Embedded Platform AIs

- **Platform Copilots:** SAP Joule, Salesforce Einstein.
- **Data-Layer AIs:** Databricks Genie, Snowflake Copilot.
- **The Catch:** Fragmented vendor silos, high dependency, and compounding licensing costs.

Core Enterprise Needs

- **Data Sovereignty:** Full control over where regulated data lives and is processed.
- **API Cost Control:** Predictable token spend and budgets at enterprise scale.
- **Security Guardrails:** One centralized policy layer governing every AI call.

How do leaders bridge the gap between rapid innovation in AI and total control?

The Sweet Spot: A Mixed Hybrid Architecture

Shift the mindset from Buy vs. Build to Buy vs. Assemble.

Leverage Existing Cloud

Utilize your foundational GCP, AWS, or Azure infrastructure for data residency.

Integrate Modular Pieces

Use lightweight, flexible orchestration tools to connect your systems.

The Result

The exact security and cost benefits of building, with the speed and reliability of buying.

The Agnostic Enterprise AI Stack

LAYER 1 — APPLICATION / UI

SAP	Salesforce	Custom Apps

LAYER 2 — ORCHESTRATION & API MANAGEMENT

LangGraph / FastAPI* (Representational)

LAYER 3 — GUARDRAILS & SECURITY

Enterprise Data Control & Compliance Layer

LAYER 4 — INFRASTRUCTURE

AWS	Azure	GCP

** Note: Components like LangGraph and FastAPI are illustrative. The architecture is fully modular and supports tools like CrewAI, Pydantic, MuleSoft, Gravitee, or A2A.*

LAYER 1: APPLICATION/ UI

Web/Mobile

Unified React/Vue frontends consuming FastAPI endpoints for enterprise-wide tasking.

Copilots

Sidecar assistants embedded in SFDC or SAP via secure plug-in architectures.

Edge Devices

Shop-floor tablets running MES-integrated diagnostic agents for operators.

LAYER 2: ORCHESTRATION & API MANAGEMENT

FastAPI Routing

Acts as the asynchronous API gateway. Handles user session ingestions, webhook triggers, and secure validation endpoints.

LangGraph Logic

Enforces predictable, stateful process loops. Standardizes complex reasoning pathways using strict state machine control.

Deterministic Paths

Restricts models to bounded workflow paths, preventing wild, non-deterministic actions on production system endpoints.

LAYER 3: SECURITY & GUARDRAILS

PII Masking & Anonymization

GCP AI Services, Microsoft Presidio and **Nvidia NeMo Guardrails** prevent proprietary formulas from leaving the network:

Scrubbing: Names, Locations, IP IDs, CAD specifications.

Hallucination Check: Cross-verifying AI outputs against Vector DB facts.

Goal Hijacking: Blocking malicious prompt injections.

LAYER 4: INFRASTRUCTURE

Leverage GCP/AWS/Azure and its AI services to deploy the Model Agnostic Gateway (Layer 2)

0%

Vendor Lock-in Risk

Commercial

Latest GPT or Gemini Models for high-reasoning financial tasks.

Open Source

Llama 3 or Mistral hosted on GCP for routine data formatting.

Multimodal

Vision models for quality inspection, package or product classification via images or factory cameras.

Appendix: Enterprise Governance & The Vulnerabilities of Vendor-Native AI

THE RISKS OF UNGOVERNED VENDOR AI

Ungoverned Vendor AI refers to the adoption of siloed, unmanaged native vendor AI features by business units without central enterprise oversight of security, data sovereignty, and cost controls.

Feature	Vendor-Native AI (Silo)	Requirements from Enterprise AI Architecture
Cross-System Context	❌ Restricted to one app	✅ SAP + SF + Dassault + Others (MES, Finance, etc.)
Security Guardrails	Vendor-controlled	Centrally Governed (Internal)
Cost Control	Variable/Opaque	Token-level Budgeting
Data Integrity	Autonomous Mutations	Human-in-the-Loop Validation

ENTERPRISE GOVERNANCE POLICY FOR VENDOR AI

Vendor AI Approved Use Cases

- **In-App Summarization:** Reading and condensing a specific MES maintenance ticket or CRM summary.
- **Local Draft Generation:** Formulating draft emails or descriptions inside the local vendor UI without execution.
- **Read-Only Research:** Sifting through native documentation isolated inside a single software silo.

Use cases controlled by Enterprise AI Org

- **Autonomous Writes:** Directly altering SQL/relational records in SAP, SF, Dassault based *solely* on text chats.
- **Cross-Boundary Actioning:** Allowing SAP AI to directly trigger shop floor execution tasks in MES.
- **Bypassing Guardrails:** Feeding proprietary formulas (Dassault PLM) into public vendor-native models.

POLICY: READ VS. MUTATE

Why Direct Writes need to be controlled by the Enterprise

Allowing non-deterministic LLMs to directly modify production databases poses an immediate threat to operational stability:

- ⚠ **Non-Determinism:** LLMs process language probabilities. A slight prompt change can mutate data unpredictably.
- ⚠ **Zero Accountability:** Actions executed by systemic service accounts strip out standard human security audits (SOX, ISO compliance).
- ⚠ **Injection Vulnerability:** Prompts can be hijacked to perform malicious queries, edits, or deletes on enterprise nodes.

THE SECURE PATTERN: DRAFT & VERIFY

1. Process Intent

The operator speaks or writes a prompt. The LLM processes natural language, mapping instructions into a validated, structured JSON schema payload.

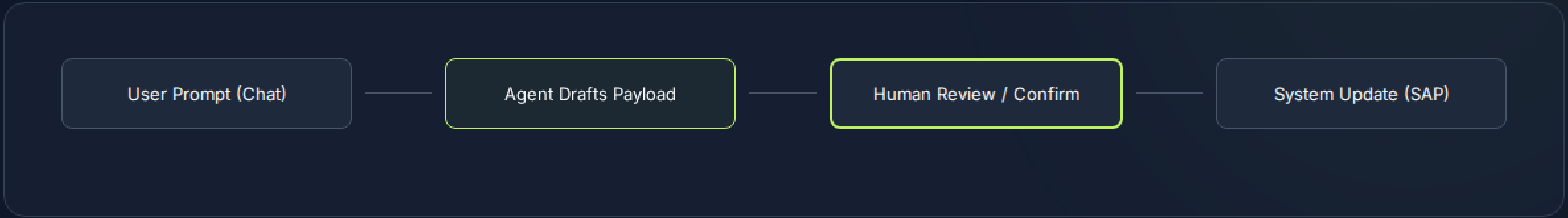
2. Populate Form

Rather than executing a database commit, FastAPI routes the JSON structure to a standard, deterministic web form on the separated UI layer.

3. Human Confirms

The operator visually reviews the populated form fields, corrects any LLM misinterpretations, and securely authorizes standard API execution.

HUMAN-IN-THE-LOOP WORKFLOW



AI creates **intent**; Humans provide **authorization**. Enterprise's main concern is to restrict non-deterministic AI that mutate production databases directly without accountability & audit trail.

Q&A, Feedback, Discussions

Manish